

Practical Exam :

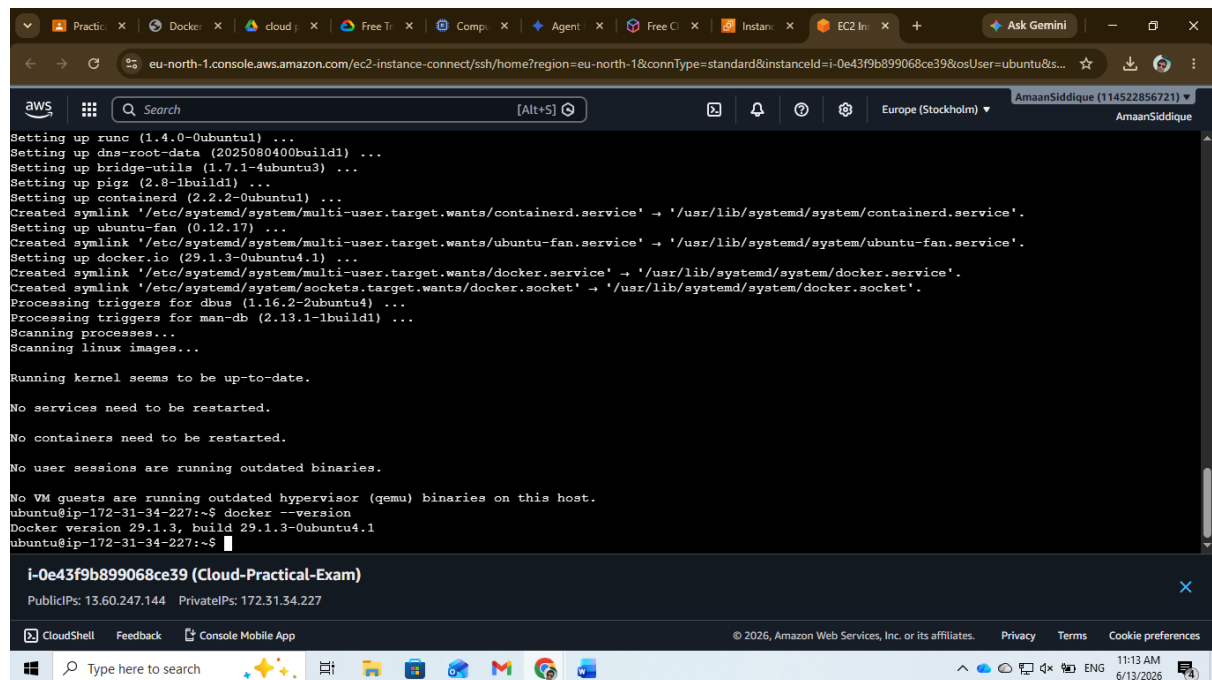
Amaan Siddique

24101A0025

Batch B

INFT

Screenshot No.1:



The screenshot shows the AWS CloudShell interface for instance i-0e43f9b899068ce39. The terminal output displays the setup of various tools (runc, dns-root-data, bridge-utils, pigz, containerd, ubuntu-fan, docker.io) and the creation of systemd symlinks for containerd, ubuntu-fan, and docker. It also shows the processing of triggers for dbus and man-db, scanning of processes and Linux images, and a confirmation that the kernel is up-to-date. The Docker version is checked as 29.1.3, build 29.1.3-0ubuntu4.1. A notification banner at the bottom indicates the instance ID and public/private IP addresses.

```
Setting up runc (1.4.0-0ubuntu1) ...
Setting up dns-root-data (2025080400build1) ...
Setting up bridge-utils (1.7.1-4ubuntu3) ...
Setting up pigz (2.8-1build1) ...
Setting up containerd (2.2.2-0ubuntu1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/containerd.service' -> '/usr/lib/systemd/system/containerd.service'.
Setting up ubuntu-fan (0.12.17) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/ubuntu-fan.service' -> '/usr/lib/systemd/system/ubuntu-fan.service'.
Setting up docker.io (29.1.3-0ubuntu4.1) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/docker.service' -> '/usr/lib/systemd/system/docker.service'.
Created symlink '/etc/systemd/system/sockets.target.wants/docker.socket' -> '/usr/lib/systemd/system/docker.socket'.
Processing triggers for dbus (1.16.2-2ubuntu4) ...
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

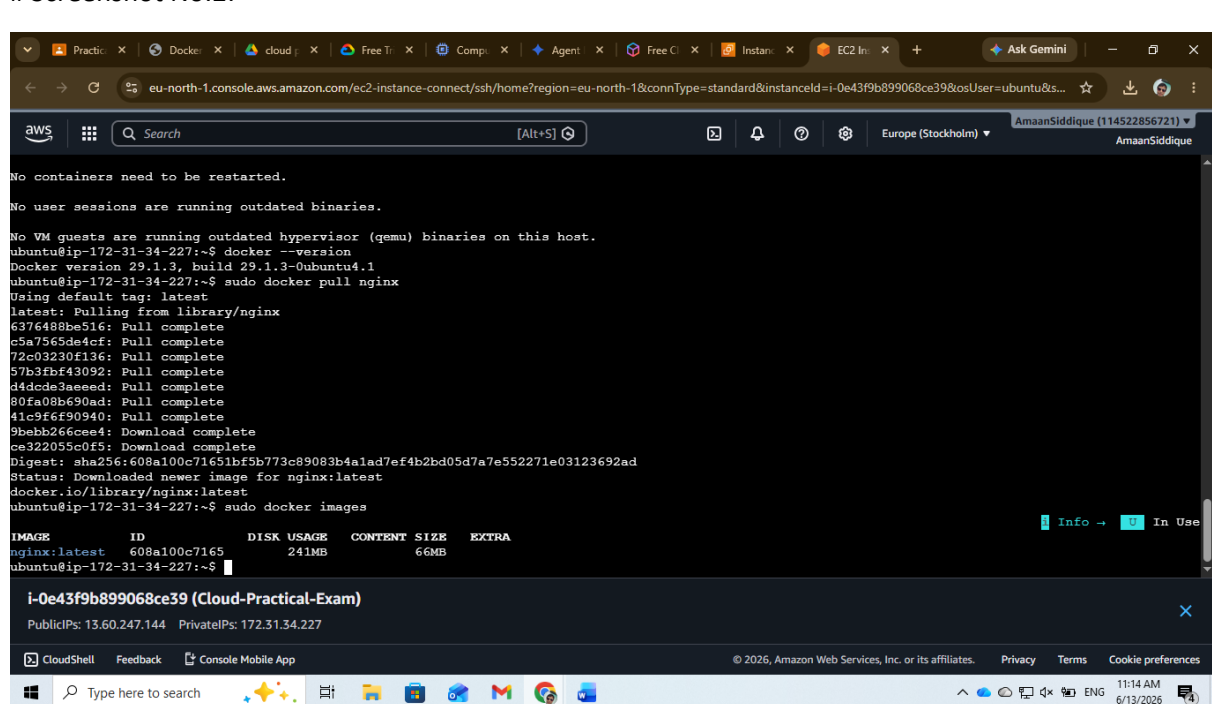
No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-34-227:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-34-227:~$
```

i-0e43f9b899068ce39 (Cloud-Practical-Exam)
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Screenshot No.2:



The screenshot shows the continuation of the Docker setup. The user runs 'docker pull nginx' and 'sudo docker images'. The terminal output shows the pull of the nginx:latest image from the library, including the digest and status. The 'docker images' command lists the pulled image. A notification banner at the bottom indicates the instance ID and public/private IP addresses.

```
No containers need to be restarted.

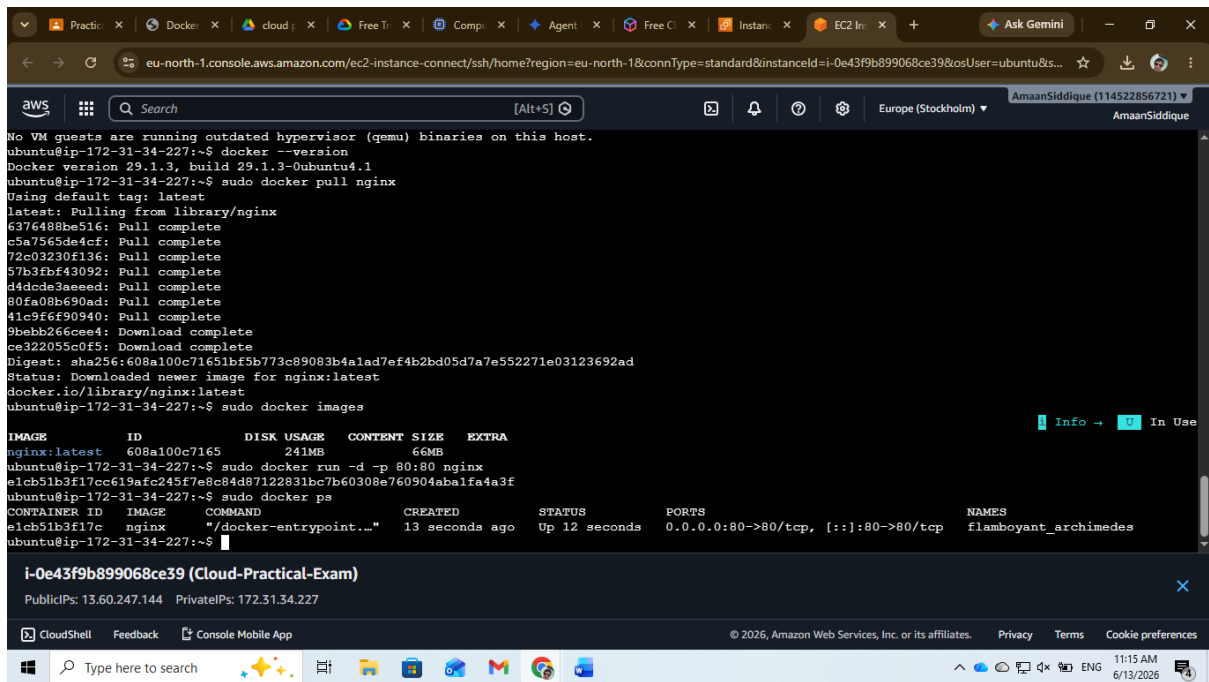
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-34-227:~$ docker --version
Docker version 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-34-227:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
6376488be516: Pull complete
c5a7565de4cf: Pull complete
72c03230f136: Pull complete
57b3bf43092: Pull complete
d4dcde3aeed: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-34-227:~$ sudo docker images
```

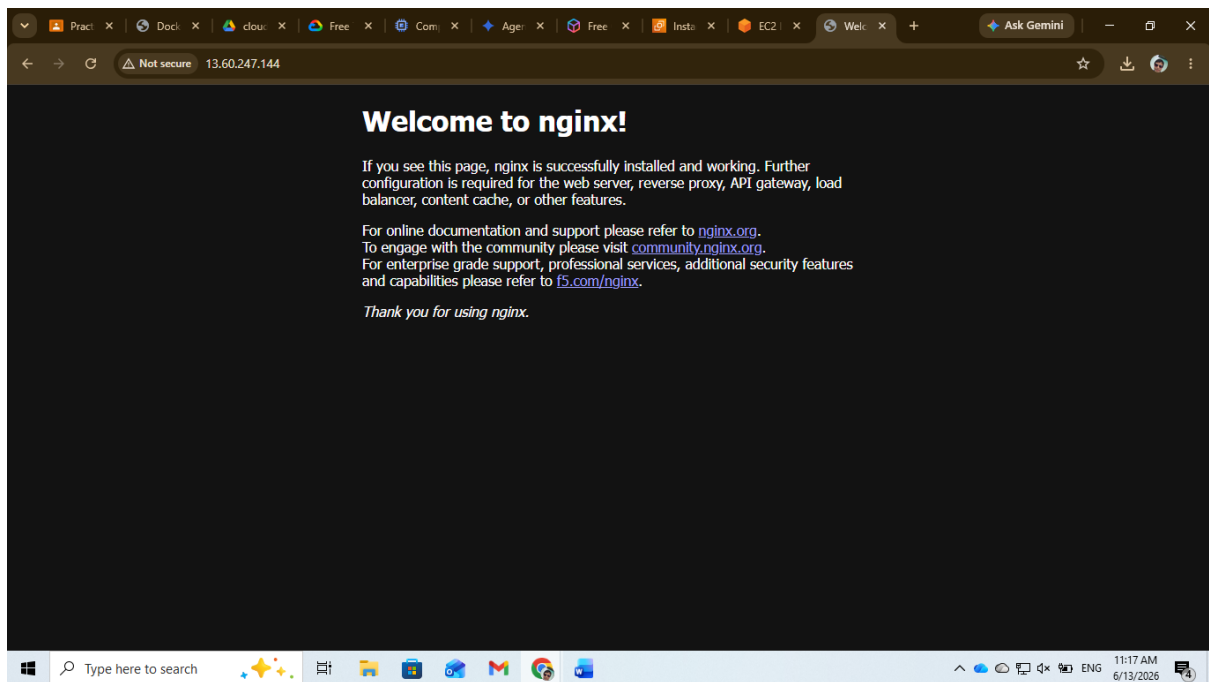
IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	608a100c7165	241MB	66MB	

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Screenshot No.3:



Screenshot No.4:



Screenshot No.5:

```
aws
[+] Search [Alt+S]
Europe (Stockholm)
AmaanSiddique (114522856721)
AmaanSiddique

Docker version: 29.1.3, build 29.1.3-0ubuntu4.1
ubuntu@ip-172-31-34-227:~$ sudo docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
6376488be516: Pull complete
c5a7565de4cf: Pull complete
72c03230f136: Pull complete
57b3fbf43092: Pull complete
d4dcde3aeeed: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-34-227:~$ sudo docker images

IMAGE          ID          DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest    608a100c7165  241MB       66MB
ubuntu@ip-172-31-34-227:~$ sudo docker run -d -p 80:80 nginx
e1cb51b3f17cc619afc245f7e8c84d87122831bc7b60308e760904abaf4a3f
ubuntu@ip-172-31-34-227:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS                  NAMES
e1cb51b3f17c   nginx     "/docker-entrypoint..." 13 seconds ago Up 12 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp flamboyant_archimedes
ubuntu@ip-172-31-34-227:~$ sudo docker volume create studentdata
studentdata
ubuntu@ip-172-31-34-227:~$
```

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Screenshot No.6:

```
aws
[+] Search [Alt+S]
Europe (Stockholm)
AmaanSiddique (114522856721)
AmaanSiddique

d4dcde3aeeed: Pull complete
80fa08b690ad: Pull complete
41c9f6f90940: Pull complete
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-34-227:~$ sudo docker images

IMAGE          ID          DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest    608a100c7165  241MB       66MB
ubuntu@ip-172-31-34-227:~$ sudo docker run -d -p 80:80 nginx
e1cb51b3f17cc619afc245f7e8c84d87122831bc7b60308e760904abaf4a3f
ubuntu@ip-172-31-34-227:~$ sudo docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED      STATUS      PORTS                  NAMES
e1cb51b3f17c   nginx     "/docker-entrypoint..." 13 seconds ago Up 12 seconds 0.0.0.0:80->80/tcp, [::]:80->80/tcp flamboyant_archimedes
ubuntu@ip-172-31-34-227:~$ sudo docker volume create studentdata
studentdata
ubuntu@ip-172-31-34-227:~$ sudo docker run -it -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
9bcbf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@4585eaae3c70:/#
```

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Screenshot No.7:

```
9bebb266cee4: Download complete
ce322055c0f5: Download complete
Digest: sha256:608a100c71651bf5b773c89083b4a1ad7ef4b2bd05d7a7e552271e03123692ad
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest
ubuntu@ip-172-31-34-227:~$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	608a100c7165	241MB	66MB	

```
ubuntu@ip-172-31-34-227:~$ sudo docker run -d -p 80:80 nginx
e1cb51b3f17cc619afc245f7e8c84d87122831bc7b60308e760904aba1fa4a3f
ubuntu@ip-172-31-34-227:~$ sudo docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e1cb51b3f17c	nginx	"/docker-entrypoint..."	13 seconds ago	Up 12 seconds	0.0.0.0:80->80/tcp, [::]:80->80/tcp	flamboyant_archimedes

```
ubuntu@ip-172-31-34-227:~$ sudo docker volume create studentdata
studentdata
ubuntu@ip-172-31-34-227:~$ sudo docker run -it -v studentdata:/data ubuntu bash
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
6f5c5aa4e145: Pull complete
1c24335ddd46: Pull complete
9bcf140d7f0f: Download complete
Digest: sha256:f3d28607ddd78734bb7f71f117f3c6706c666b8b76cbff7c9ff6e5718d46ff64
Status: Downloaded newer image for ubuntu:latest
root@4585eaae3c70:/# echo "Name: Amaan Siddique, Roll Number: 24101A0025, CDAC Batch B" > /data/info.txt
root@4585eaae3c70:/# cat /data/info.txt
Name: Amaan Siddique, Roll Number: 24101A0025, CDAC Batch B
root@4585eaae3c70:/#
```

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Screenshot No.8:

```
root@4585eaae3c70:/# cat /data/info.txt
Name: Amaan Siddique, Roll Number: 24101A0025, CDAC Batch B
root@4585eaae3c70:/# exit
exit
ubuntu@ip-172-31-34-227:~$ mkdir exam_q3
ubuntu@ip-172-31-34-227:~$ cd exam_q3
ubuntu@ip-172-31-34-227:~/exam_q3$ nano Dockerfile
ubuntu@ip-172-31-34-227:~/exam_q3$ sudo docker build -t cloud-exam-image .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

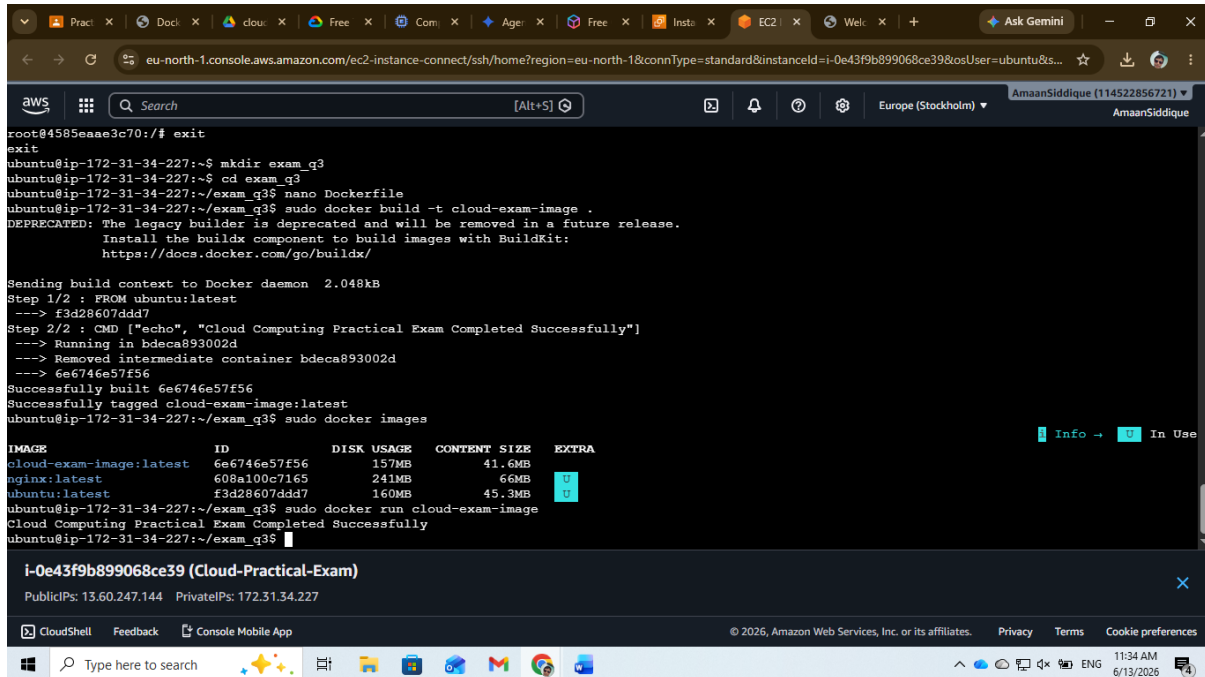
Sending build context to Docker daemon 2.048kB
Step 1/2 : FROM ubuntu:latest
--> f3d28607ddd7
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in bdeca893002d
--> Removed intermediate container bdeca893002d
--> 6e6746e57f56
Successfully built 6e6746e57f56
Successfully tagged cloud-exam-image:latest
ubuntu@ip-172-31-34-227:~/exam_q3$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
cloud-exam-image:latest	6e6746e57f56	157MB	41.6MB	
nginx:latest	608a100c7165	241MB	66MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	U

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Screenshot No.9:



The screenshot shows an AWS CloudShell terminal window with the following content:

```
root@i-0e43f9b899068ce39:~# exit
exit
ubuntu@ip-172-31-34-227:~$ mkdir exam_q3
ubuntu@ip-172-31-34-227:~$ cd exam_q3
ubuntu@ip-172-31-34-227:~/exam_q3$ nano Dockerfile
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  2.048kB
Step 1/2 : FROM ubuntu:latest
--> f3d28607ddd7
Step 2/2 : CMD ["echo", "Cloud Computing Practical Exam Completed Successfully"]
--> Running in bdeca893002d
--> Removed intermediate container bdeca893002d
--> 6e6746e57f56
Successfully built 6e6746e57f56
Successfully tagged cloud-exam-image:latest
ubuntu@ip-172-31-34-227:~/exam_q3$ sudo docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
cloud-exam-image:latest	6e6746e57f56	157MB	41.6MB	
nginx:latest	608a100c7165	241MB	66MB	U
ubuntu:latest	f3d28607ddd7	160MB	45.3MB	U

```
ubuntu@ip-172-31-34-227:~/exam_q3$ sudo docker run cloud-exam-image
Cloud Computing Practical Exam Completed Successfully
ubuntu@ip-172-31-34-227:~/exam_q3$
```

Below the terminal output, a summary box for the instance **i-0e43f9b899068ce39 (Cloud-Practical-Exam)** is visible, showing Public IPs: 13.60.247.144 and Private IPs: 172.31.34.227.

Bonus Challenge (Optional):

1. Difference between a Docker Image and a Docker Container

A Docker Image is a read-only, static template containing the instructions and code needed to run an application. A Docker Container is the live, running instance of that image.

2. Why are Docker Volumes used?

Docker volumes are used to permanently save data generated by a container so it isn't lost when the container is deleted or stopped. They also allow multiple containers to share the same data safely.